UFMG
UNIVERSIDADE FEDERAL
DE MINAS GERAIS

Location–Routing for the last-mile problem

bringing stationary and mobile parcel self-serviced smart lockers to underserved remote areas of Minas Gerais, Brazil

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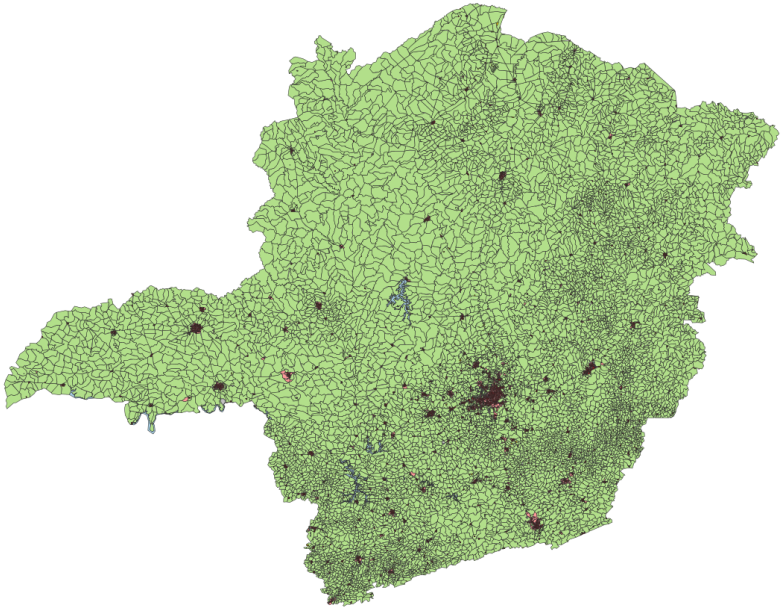
desigining a network of stationary and mobile parcel self-serviced smart lockers to maximize the captured market share while taking into account current competitors.

e-Commerce of Minas Gerais, Brazil, and its challenges

current status:

- 2nd largest e-commerce of Brazil
- revenue 2024 $\approx$ US$ 6 billions
- delivered volume increased 45% 2025



current challenges:

- pressure for new delivery services
- need for new macro and micro hubs
- awareness of competitors' movements
- compete or cooperate? but how?
- how to model and solve this?
- countryside of MG: $\approx$ 20,000 points of delivery